



Paper id: 250604

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Subject Code: RCAI604

Roll No:

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MCA-INT
(SEM VI) THEORY EXAMINATION 2024-25
INTRODUCTION TO AUTOMATA THEORY & LANGUAGES

TIME: 3 HRS

M.MARKS: 70

Note: Attempt all Sections. In case of any missing data; choose suitably.

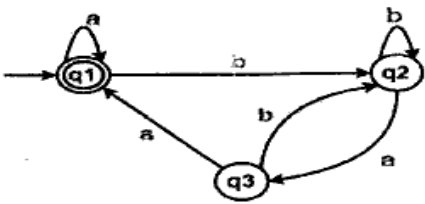
SECTION A1. Attempt *all* questions in brief.

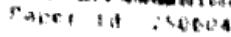
02 x 7 = 14

Q no.	Question	CO	Level
a.	What are the differences between DFA and NDFA?	1	K2
b.	Design a NFA that accepts all the strings for input alphabet $\Sigma = \{a,b\}$ containing the substring abba.	1	K2
c.	What is ϵ -closure? Give example.	1	K2
d.	What are the different operations performed on regular languages?	1	K2
e.	Write regular expression for set of all strings such that no two zeros come together over $\Sigma = \{0,1\}$.	2	K3
f.	What are the differences between Mealy and Moore Machine?	1	K2
g.	Explain Recursively enumerable language.	5	K3

SECTION B2. Attempt any *three* of the following:

07 x 3 = 21

Q no.	Question	CO	Level
a.	Find regular expression for the following DFA using Arden's Theorem. State and Prove Arden's theorem. 	1	K2
b.	Convert the following CFG to its equivalent GNF (Greibach Normal Form): $S \rightarrow AA \mid a, A \rightarrow SS \mid b$	3	K3
c.	What is pumping lemma for regular languages? State and prove it, also prove that $L = \{a^n b^n : n > 0\}$ is not regular.		
d.	Convert the following CFG into CNF (Chomsky Normal form): $S \rightarrow aAa \mid bSb \mid \lambda, A \rightarrow C \mid a, B \rightarrow C \mid b, C \rightarrow CDE \mid \lambda, D \rightarrow A \mid B \mid ab$	3	K3



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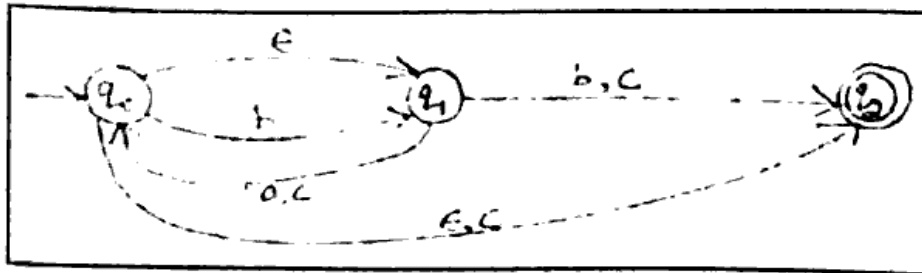
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Construct a minimum state DFA from given FA



SECTION C

3. Attempt any *one* part of the following:

07 x 1 = 07

Q no.	Question	CO	Level
a.	Construct the minimum state automata equivalent to DFA described by following Fig: https://www.aktuonline.com	1	K3
b.	Explain the Closure properties of regular languages.	2	K3

4. Attempt any *one* part of the following:

$$07 \times 1 = 07$$

Q no.	Question	CO	Level
a.	Design a Moore and Mealy machine for a binary input sequence such that if it has a substring 101, the machine outputs A if input has substring 110 it outputs B otherwise it outputs C.	1	K3
b.	Show that $L = \{ww \mid w \in \{a,b\}^*\}$ is not regular.	2	K3

5. Attempt any *one* part of the following:

$$07 \times 1 = 07$$

Q no.	Question	CO	Level
a.	Design a PDA for the Language $L = \{WW^R \mid W = \{a, b\}^*\}$	3	K3
b.	Design a Turing machine that copies strings of 1's, for any $w \in \{1\}^*$	5	K3



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(SEM VI) THEORY EXAMINATION 2024-25

INTRODUCTION TO AUTOMATA THEORY & LANGUAGES

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M.MARKS: 70

6. Attempt any *one* part of the following:

07 x 1 = 07

Q no.	Question	CO	Level
a.	Prove that the following language $L = \{ a^n b^n c^n \}$ is not Context Free.	3	K3
b.	Design DFA which accepts all strings for a binary number divisible by 3.	1	K3

7. Attempt any *one* part of the following:

07 x 1 = 07

Q no.	Question	CO	Level
a.	Write short notes on the following: (i) Recursive Language and Recursively Enumerable Language. (ii) PCP problem and Modified PCP Problem	5	K3
b.	Design a TM for the following language $L = \{ a^n b^n c^n n \geq 1 \}$	5	K3